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EGCNet: a hierarchical graph convolutional neural network for improved classification of electrocardiograms

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Abstract

The automatic classification of electrocardiograms (ECGs) plays a crucial role in the early diagnosis of cardiovascular diseases. In recent research, deep neural network (DNN)-based methods have garnered significant attention due to their exceptional feature extraction capabilities. However, these methods face challenges in dealing with the complex inter- and intra-class dependencies inherent in different arrhythmias. We propose a classification model for electrocardiograms based on the Graph Convolutional Neural Network (EGCNet) to address this. In our EGCNet framework, we first employ the row and column 1-dimensional residual convolution to learn intra-class dependency. Then, we construct a hierarchical graph structure to model the inter-class dependency. In such a graph, each node represents one type of disease, and the edges describe the relationship between diseases. To reduce the complexity of inter-class dependency, we build a hierarchical graph structure by treating similar nodes as the same type. Thus, we can learn the discriminant dependency in the higher-level graph space and feed this high-level knowledge back to the lower-level graph space. Finally, we evaluated our method using publicly available datasets from the PhysioNet/Computing Challenge 2020 ECG data. For instance, our model achieved robust performance in the 24-label experiment with a precision of 0.753 and an F1 score of 0.722.

Keywords: ECG arrhythmia classification, Graph convolution, Inter- and intra-class dependency, Deep learning, Graph Pooling

1 Introduction

1.1 Motivation

Accurate classification of arrhythmias from the electrocardiograms (ECGs) is essential for patients with cardiovascular diseases [1]. Arrhythmias encompass various cardiac rhythm disorders with distinct characteristics and clinical implications. According to ECG arrhythmias interpretation guidelines [2], more than 60 arrhythmias need to be identified in the clinic. Among them, different types of arrhythmias can lead to very different consequences [3, 4]. For example, asymptomatic sinus arrhythmia has no

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impact on the patients' health, and there is no need for further treatment. On the contrary, ventricular tachycardia can cause cardiac arrest and sudden death, which needs timely medical treatment or even surgery [5]. This means if we can subdivide each type of arrhythmia, it can greatly improve the efficiency of clinical diagnosis and treatment to improve patients' health. However, in clinical practice, this goal is difficult to achieve. On the one hand, the analysis of the ECGs mainly depended on highly trained and skilled cardiologists, which is expensive and time-consuming. The efforts [6, 7] based on automated methods focus only on a limited number of arrhythmias but lack full identification of all arrhythmia types. To this end, it is urgent to develop an automatic method to accurately identify arrhythmias.

However, the full classification of all arrhythmias is still challenging due to the complex dependency in arrhythmias [8]. This dependency can summarized into two aspects. 1) Inter-class dependency: A patient may suffer from one or more arrhythmias at the same time. The relationship between various arrhythmias may be highly correlated, mutually exclusive, or irrelevant, which increases the difficulty of classifying arrhythmias. 2) Intra-class dependency: According to the data characteristics in dimension and time sequences, this dependency can be further divided into two parts. The first is cross-lead dependency; the relationship between different leads is also important to distinguish arrhythmia. The second is long-term dependency: this dependence refers to the context among several beats over a long period, which is more difficult to address than the context of the waveform inside a beat.

1.2 Related works

1.2.1 Traditional machine learning methods

While traditional machine learning methods based on hand-crafted ECG features [9–12] face challenges in addressing the mentioned issues, it is important to acknowledge that in certain cases, hand-crafted features can achieve outstanding results in arrhythmia classification. For example, time-domain analysis features, such as those using RR-intervals for heart rate variability assessment (arrhythmia identification), can yield good classification results without complex and expensive computations [13–16]. These methods rely on human analysis of ECG curves, including auto-correlation function, frequency domain features, wavelet transform, higher-order statistics (HOS), and morphological features (such as interval between beats, amplitude, area). However, the classification accuracy of hand-crafted features often depends on the quality of the features. If the quality of the extracted features is low-quality, it is difficult to achieve good classification results even with a well-designed classifier [16]. Additionally, hand-crafted features are often insufficiently tolerant to noise. When the ECG measurement equipment changes, the performance of manual features-based methods may decrease [17]. Thus, while hand-crafted features can be effective in certain scenarios, automatic feature extraction methods are generally more advantageous in handling the complex and diverse dependencies inherent in different arrhythmias.

1.2.2 Deep learning methods

Deep learning methods have greatly improved the diagnosis of cardiac arrhythmias. However, most of these studies have focused on limited cardiac rhythm abnormalities

[18–23] without considering the complex dependency among a large number of the arrhythmias. Convolutional neural networks (CNNs) are the commonly used framework to adaptively extract hierarchical features from the data [24]. In order to adapt to 2-dimensional CNNs, many methods [25–28] convert a 1-dimensional ECGs sequence into a 2-dimensional spectrogram and then extract the features by the CNNs. However, converting a 1-dimensional sequence to a two-dimensional spectrum brings additional computational costs. Besides, compressing the feature dimension will also lead to losing information on the original signal. Therefore, the other studies try to avoid this by learning features directly from 1-dimensional signals. In this case, 1-dimensional CNNs [29–32] and recurrent neural networks(RNNs) [33–36] are naturally employed for 1-dimensional ECGs learning. Compared to the 1-dimensional CNNs, the RNNs show more powerful performance in analyzing time series data. Following a data-driven paradigm, the RNNs-based model can track the hidden state encoding dynamic historical information through the recurrent connection structure. Many works have applied RNNs [33] and its variant models [34, 37] to arrhythmia diagnosis and analysis and achieved promising results. However, RNNs may ignore the context due to the long period of the ECG signals. These mentioned deep neural network-based methods have progressed greatly compared to traditional machine learning algorithms. However, most of these works [20, 21, 38–42] concentrate on a limited number of arrhythmias. In this way, although they can achieve good accuracy, their clinical value is greatly limited by the lower accuracy and insufficient recognition of types of arrhythmias.

Recently, the work [43] proposes a multi-label classification paradigm to solve the diagnosis of arrhythmias, in which the dependency among abnormalities is modeled. In this method, a CNN is firstly employed to map raw ECG data into different feature spaces; then, a deep graph network is employed to model the relationship between arrhythmias. However, this method only considers the relationship between abnormalities within a single layer. 1) Since the wide variety of arrhythmias, many relationships between abnormalities need to be learned. In this case, the relationship between each pair of arrhythmias is undoubtedly complex and difficult to converge. Besides, 2) during the feature updating process, the model may pay too much attention to the dense relationship in most classes while ignoring the sparse relationship in the minority classes. Therefore, it is still challenging to accurately learn the relationship between abnormalities.

1.3 Contributions

In this paper, we propose a hierarchical graph network for classifying arrhythmia. Figure 1 illustrates the general process of ECG classification and the similarities and differences among various ECG classification methodologies. Figure 1a elucidates that automatic ECG classification aims to alleviate the burden of screening various anomaly types on physicians and enhance the efficiency of clinical diagnosis and treatment. Figure 1b summarizes the diverse mainstream automatic ECG classification algorithms discussed earlier, providing a consolidated overview of the prevailing methodologies in this domain. Our model consists of one signal feature extraction module and one graph learning module. In the former, the long-term dependency of the ECG is extracted by the multi-scale 1-dimensional CNNs. Subsequently, we introduce a perceptron network to address the cross-head dependencies, which refer

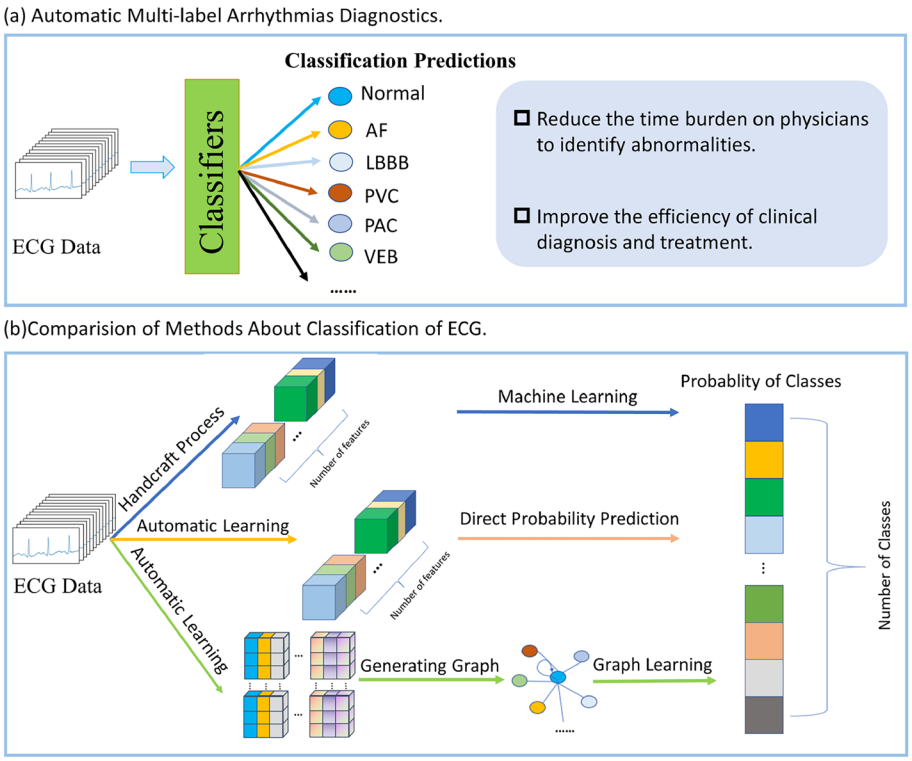


Fig. 1 **a** Automatic multi-label arrhythmias diagnostics. **b** Comparison of methods about classification of ECG

to the intricate relationships between different ECG leads. This network functions as an intermediary, fostering communication and interaction between the features extracted from various leads, ensuring a comprehensive and integrated analysis. In the latter, we employ hierarchical graph neural networks to address the inter-class dependency in different layers. In the graph network, we use the vertexes to represent the arrhythmia features, while the edges represent the relationship among them. There are two learning steps in our hierarchical graph neural networks: 1) we first construct a hierarchical graph via the vertex infomax pooling (VIPool); 2) we update the feature representations via the graph convolutional operation in each graph layer.

Our advantages can be summarized as follows:

1 We propose a bidirectional feature extraction mechanism to learn the intra-class dependency, effectively capturing both temporal dependence and cross-head dependence of raw ECG signals. This mechanism significantly enhances feature extraction and improves classification accuracy.
2 We employ a hierarchical graph neural network to learn the inter-class dependency. Our model effectively captures complex inter-class relationships through multi-scale graph modeling and prevents local graph structure damage via vertex infomax pooling, leading to robust classification performance.
3 Our model achieved strong performance results, particularly in the 24-label experiment, where it attained a precision of 0.753 and an F1 score of 0.722, demonstrating

its effectiveness in accurate arrhythmia classification across a wide range of scenarios.

2 Materials And Methods

2.1 MODEL OVERVIEW

The overall framework of the model is shown in Fig. 2, which includes the preprocessing part for the ECG signal, the bidirectional feature extraction network for intra-class dependency learning, and the hierarchical graph network for inter-class dependency Learning. The functions of each part will be described in detail later.

2.2 Data Preprocessing

The ECG recordings we have collected exhibit disparities in acquisition frequencies and durations due to the variability in equipment across different hospitals. Such discrepancies, if unaddressed, could potentially introduce biases and negatively impact the performance of deep neural network models trained on these data. To mitigate these issues and ensure consistency in our dataset, we have standardized the sampling frequencies and lengths of all ECG signals, aligning them with a common framework before training our deep neural network. We employ the fast Fourier transform [44] to adjust the sampling frequency. Assuming the original signal is x, it can be calculated by the fast Fourier transform:

$$y_j = \sum_{k=1}^{n} x_k * \mathrm{e}^{\frac{2\mathrm{ki}\pi}{\mathrm{n}}} \tag{1}$$

where y_j represents the j-th element of the FFT output, n denotes the total number of data points, x_k corresponds to the k-th element of the input ECG signal, $e^{2\pi i \cdot k \cdot j/n}$ signifies the complex exponential component essential for FFT computation.

A critical component of our analysis involves addressing variations in sampling frequencies and lengths among different ECG recordings, which often result from using diverse equipment across various hospitals and data sources. To ensure data uniformity,

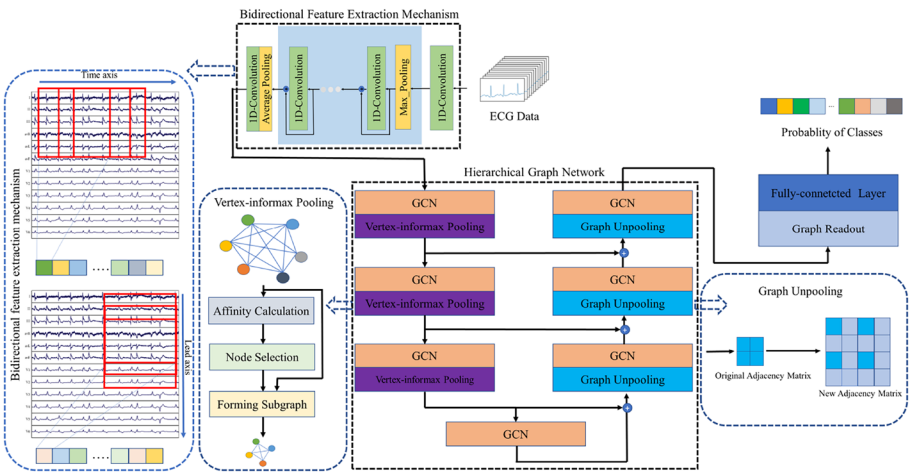


Fig. 2 Architecture of our proposed EGCNet. Our model contains three parts: graph representation generating module, graph learning module, and graph readout and classification module

which is pivotal for the accuracy of FFT-based frequency analysis, we standardize the sampling frequency of each ECG signal to match a predefined common standard. This standardization process includes techniques such as resampling and interpolation, tailored specifically to the characteristics of each data source.

Similarly, to maintain consistency across the dataset, we adjust the lengths of the ECG signals using methods such as truncation or padding. This ensures all recordings are of uniform length prior to FFT application, thereby eliminating any discrepancies that could affect the subsequent analysis.

These preprocessing steps are essential for normalizing the ECG data, thereby enhancing the reliability and consistency of our findings. By methodically adjusting both the frequency and length of the ECG recordings, we prepare the dataset thoroughly for detailed FFT analysis, ensuring that our results are robust and replicable across various equipment and data sources.

2.3 Bidirectional Feature Extraction Mechanism for Intra-class Dependency Learning

We propose a bidirectional feature extraction mechanism to address both the temporal and cross-head dependence. Specifically, we employ two one-dimensional networks to extract the feature from the temporal and cross-head directions. As shown in Fig. 2, in the temporal direction, we set the size of the first convolution kernel as (1*15) to expand its receptive field. Simultaneously, we incorporate an additional convolutional kernel (1*7) to capture information at multiple scales, encompassing variations across different leads and time points. In the cross-head direction, we set the convolution kernel as (1*5) to address the dependence across 12 heads of ECG signals. The residual architecture, as demonstrated in ResNet34 [45], has proved to be highly effective in capturing intricate patterns and dependencies within data. Given the complexity and hierarchical nature of ECG signals, we sought to leverage this proven approach to enhance our model's capability in feature extraction. Using four residual blocks, we aim to empower our model to learn and represent both local and global features of ECG data, ultimately contributing to improved classification performance. Then, the average pooling is performed on the output variables from the last block. Subsequently, we obtain a feature map X, defined as $X = \{x_1, x_2, ..., x_n\}$, where each x_i represents the feature expression of the i-th arrhythmia derived from our bidirectional feature extraction mechanism. We can map the fully connected layer into n feature spaces corresponding to different arrhythmias.

feature extraction mechanism.

2.4 Hierarchical Graph Network for Inter-class Dependency Learning

As mentioned above, the feature expression of different arrhythmias is obtained by a bidirectional feature extraction mechanism. Combined with relationships among those features, we can employ a graph $G(\mathcal{V}, \mathbf{A}, \mathbf{W}, \mathbf{X})$ to model the inter-class dependency between different arrhythmias. $\mathcal{V} = \{v_1, \ldots, v_N\}$ is the set of vertices of graph G, where each vertex represents on type of arrhythmia; A is the adjacency matrix of graph G, which describes the relationship between different arrhythmias; W is the weight matrix assigned to the edges; X is the feature matrix of the nodes.

The graph G has built dependencies between different arrhythmias. However, it is difficult to learn those complex dependencies. This difficulty is reflected in two aspects: 1) The relationships between all diseases are considered simultaneously, which leads to a lot of edges (relationships) in the graph that need to be learned; 2) a patient may suffer from one or more types of arrhythmia at the same time, resulting in a variable weight of the relationship. To address those difficulties, we proposed to cluster similar nodes to construct a super-graph. In such a super-graph, we can focus on some key relationships. We first introduce the graph convolution layer for feature learning in the next two subsections. Then, we introduce vertex infomax pooling for hierarchical graph construction.

2.4.1 Graph Convolution Layer

In this subsection, graph convolution network (GCN) is used to update the feature matrix and the edges of the disease nodes [46]. Following the bidirectional feature extraction mechanism, we can acquire a graph $G(V, A, \mathbf{W}, \mathbf{X})$. Subsequently, the Laplace matrix, defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$ with $\mathbf{D} \in \mathbb{R}^{N\times N}$ being a diagonal matrix, can be decomposed into:

$$\mathbf{L} = \mathbf{U}\Lambda\mathbf{U}^T \tag{2}$$

where $\mathbf{U} = (\mathbf{u}_1, \ldots \mathbf{u}_l, \ldots \mathbf{u}_n)$ contains a set of orthogonal eigenvectors $\mathbf{U}\mathbf{U}^T = \mathbf{I}$. And Λ is the eigenvalue matrix, which can be expressed as $\Lambda = \mathrm{diag}(\lambda_1, \ldots \lambda_l, \ldots \lambda_n)$.

The graph convolution is derived from the theory of spectral convolution. The graph convolution can be expressed as:

$$\mathbf{x} * \mathbf{h} = \mathbf{U}h(\Lambda)\mathbf{U}^{\mathrm{T}}\mathbf{x} \tag{3}$$

Among them, x represents the feature vector of the graph, which is the deep feature representation of all diseases. $h(\Lambda)$ is a diagonal matrix, which is obtained by transforming the eigenvectors.

To alleviate the computational burden associated with graph convolution, Hammond et al. [47] introduced an approximation technique utilizing Chebyshev polynomials to represent the convolution kernel. This approach significantly simplifies the convolution operations. Specifically, the function $h(\Lambda)$ is approximated as follows:

$$h(\Lambda) \approx \sum_{k=0}^{K} \theta_k T_k(\widetilde{\Lambda}), \tag{4}$$

where θ_k denotes the coefficient corresponding to the k-th Chebyshev polynomial, and $T_k(\widetilde{\Lambda})$ represents the k-th Chebyshev polynomial evaluated at $\widetilde{\Lambda} = \frac{2\Lambda}{\lambda_{\max}} - I$. In this context, I stands for the identity matrix, K is the order up to which the Chebyshev polynomials are considered, and $\lambda_{\max}$ is the maximum eigenvalue of Λ.

In the GCN layer, we introduce two simplifying assumptions to facilitate the model's computational efficiency: we approximate the maximum eigenvalue $\lambda_{\max}$ as 2, and set the Chebyshev polynomial's order K to 1. These assumptions significantly streamline the model.

To further reduce the parameter count, we approximate θ_0 and θ_1 such that $\theta = \theta_0 = -\theta_1$. Consequently, the graph convolution update process simplifies to:

$$Ug_\theta(\Lambda)U^\top x \approx \theta\left(I + D^{-\frac{1}{2}}AD^{-\frac{1}{2}}\right)x \tag{5}$$

Additionally, considering the relationships among nodes, the GCN update can be alternatively expressed as:

$$\begin{aligned} h_i^{(l+1)} &= \sigma\left(b^{(l)} + \sum_{j\in\mathcal{N}(i)} \frac{e_{ji}}{c_{ji}} h_j^{(l)} H^{(l)}\right), \\ c_{ji} &= \sqrt{|\mathcal{N}(j)|}\sqrt{|\mathcal{N}(i)|}, \end{aligned} \tag{6}$$

where l denotes the layer number, $b^{(l)}$ is the bias, i and j are nodes, $\mathcal{N}(i)$ represents the neighboring nodes of node i, c_{ji} is the product of the square roots of the degrees of nodes i and j, $H^{(l)}$ denotes the learnable parameters of the l-th layer, $h_j^{(l)}$ is the feature vector of node j, and e_{ji} is the edge weight between nodes i and j.

2.4.2 Vertex Informax Pooling

We have a basic assumption that if a node can fully represent its neighbor nodes, then this node can be regarded as a super node in the super-graph. In other words, we can ignore its neighbor nodes to simplify the graph structure. Therefore, the model can pay more attention to the relationship between these super nodes. We employ the vertex informax pool (VIPool) to measure the representational ability of each node. The VIPool maintains vertices with the highest mutual information with its surrounding neighbors. In the graph $G(\mathcal{V}, \mathbf{A}, \mathbf{W}, \mathbf{X})$, the goal of pooling is to find an optimal subset of K nodes; this process can be expressed as:

$$\max_{\Omega\subset\mathcal{V}} C(\Omega), \text{ subject to } |\Omega| = K \tag{7}$$

where $C(\Omega)$ represents the quantization function of mutual information between the node and its neighbor nodes, K represents the number of selected subset nodes, and Ω is the set of subset nodes, $|\Omega|$ represents the number of nodes in the subgraph.

In our model, KL divergence [48] is introduced as the mutual information between a node and its neighboring nodes. Assuming that a node $v$is randomly selected from graph G, N_v is the set of neighbor nodes of the v, the feature of the node v can be expressed as $P_{\mathbf{v}} = P(\mathbf{v} = \mathbf{x}_v)$, where x_v represents the feature of vertex v. Similarly, the feature value distribution of neighbor nodes of the node v can also be expressed as $P_{\mathbf{n}} = P\left(\mathbf{n} = \mathbf{y}_{\mathcal{N}_u}\right)$, where y_{N_u} is the features of vertex u's neighborhood. Therefore, the mutual information value of node v and its neighbor nodes can be expressed as:

$$I^{(\Omega)}(\mathbf{v}, \mathbf{n}) = D_{\mathrm{KL}}\left(P_{\mathbf{v},\mathbf{n}} \| P_{\mathbf{v}} \otimes P_{\mathbf{n}}\right) \tag{8}$$

According to the relationship between f divergence [49] and KL divergence, we then get:

$$\begin{aligned} I^{(\Omega)}(\mathbf{v}, \mathbf{n}) \geq \sup_{T\in\mathcal{T}} \{&\mathbb{E}_{\mathbf{x}_v, \mathbf{y}_{\mathcal{N}_v}\sim P_{\mathbf{v},\mathbf{n}}}\left[T\left(\mathbf{x}_v, \mathbf{y}_{\mathcal{N}_v}\right)\right] - \\ &\mathbb{E}_{\mathbf{x}_v\sim P_{\mathbf{v}}, \mathbf{y}_{\mathcal{N}_u}\sim P_{\mathbf{n}}}\left[e^{T(\mathbf{x}_v, \mathbf{y}_{\mathcal{N}_u})-1}\right] \end{aligned} \tag{9}$$

Among them, $T \in \mathcal{T}$ is any function that can map a pair of nodes v and neighbor nodes to real values. $I^{(\Omega)}(\mathbf{v}, \mathbf{n})$ denotes the mutual information between a node v and its neighborhood nodes n. x_v, and $y_{(N_v)}$ are the features of the node v and its neighbors, respectively, drawn from the joint distribution $P_{v,n}$. The first expectation, $\mathbb{E}_{x_v, y_{N_v} \sim P_{v,n}}\left[T\left(x_v, y_{N_v}\right)\right]$, calculates the expected value of T under the joint distribution of v and n, capturing their mutual dependency. The second expectation, $\log\left(\mathbb{E}_{x_v \sim P_v, y_{Nu} \sim P_n}\left[e^{T(x_v, y_{N_u})}\right]\right)$, represents the expected exponential of T under the product of the marginal distributions of v and n, serving as a normalization term.

Neural network is used to replace the mapping function, so the objective function that needs to be optimized becomes:

$$C(\Omega) = \max_{w} \frac{1}{|\Omega|} \sum_{v \in \Omega} \log \sigma\left(T_w\left(\mathbf{x}_v, \mathbf{y}_{\mathcal{N}_v}\right)\right) + \frac{1}{|\Omega|^2} \sum_{(v,u) \in \Omega} \log\left(1 - \sigma\left(T_w\left(\boldsymbol{x}_v, \boldsymbol{y}_{\mathcal{N}_u}\right)\right)\right) \tag{10}$$

Among them, $C(\Omega)$ aims to maximize the mutual information between each node v and its neighborhood through the neural network T_w. σ denotes the sigmoid function, transforming the output of T_w into a probability between 0 and 1. The first term inside the maximization encourages high mutual information between v and its actual neighbors N_v, pushing T_w to produce higher values for true neighbor pairs. The second term penalizes high mutual information estimates between v and all other nodes N_u not in its immediate neighborhood, encouraging T_w to differentiate between actual neighbors and non-neighbors. This objective function facilitates the learning of node representations that preserve and maximize the essential information content of the graph structure, enhancing our model's ability to capture and utilize the complex relationships inherent in the data.

Unpooling needs to use the previously recorded node index and generate a new graph feature matrix $\mathbf{X}' = \mathbf{0} \in \{0\}^{N \times d}$ and is used to restore the reduced graph to its original shape and add the feature matrix from the same layer.

2.5 Graph Readout and Classification

In the final, we employ a READOUT layer for the graph classification. After hierarchical graph learning, we can get a feature set of all nodes $(X_v | v \in G)$; then, all nodes on of graph are converted into the final classification of the arrhythmias:

$$X_G = \mathrm{READOUT}\left(\left\{X_v^{(K)} \mid v \in G\right\}\right) \tag{11}$$

Among them, the READOUT module? represents the calculation method of mean, maximum, minimum, and sum, respectively. Finally, according to the features of the entire graph obtained from the graph READOUT?, it is mapped through the fully connected layer to arrhythmias that need to be predicted.

2.6 Loss Function

Due to the datasets we used, the number of samples included in each type of arrhythmia is unbalanced. A weighted loss function is utilized to increase the performance of model. The weight corresponding to each type of arrhythmia is related to its number in datasets.

$$\begin{aligned} \ell(\hat{y}, y) = L &= \sum_{n=1}^{N} l_n \\ l_n &= -w_n \left[y_n \cdot \log \hat{y}_n + \left(1 - y_n\right) \cdot \log \left(1 - \hat{y}_n\right) \right] \\ w_n &= \frac{1}{\log \left(\sum_i^n \mathrm{y_i}\right)} \end{aligned} \tag{12}$$

where w_n is the weight corresponding to various ECG abnormalities, $\hat{y}_n$is the probability predicted by the network, y_n is the label value, and the value is {0, 1}.

2.7 Implementation of EGCNet

Algorithm 1 Hierarchical Graph Convolutional Network Classification for ECG Signals

Require: Set of raw ECG signals $\mathcal{S} = \{S_1, S_2, \ldots, S_N\}$
Ensure: Classification labels for each ECG signal
Step 1: Preprocess ECG Signals
for each signal $S_i \in \mathcal{S}$ **do**
    Standardize sampling frequency and length
    Apply FFT to adjust sampling frequency
    Normalize to zero mean and unit variance
end for
Step 2: Extract Features
for each preprocessed signal S_i **do**
    Extract temporal and spatial features using 1D CNN
    Enhance features using residual blocks
end for
Step 3: Construct Hierarchical Graph
Initialize graph $G(V, E)$ with vertices V representing signals
Apply Vertex-Infomax Pooling (VIPool) to cluster nodes
Step 4: Apply Graph Convolution
for each layer in the hierarchical graph **do**
    for each node $v \in V$ **do**
        Update v's feature using GCN based on neighboring nodes
    end for
end for
Step 5: Classify ECG Signals
Aggregate node features using a READOUT function
Map to classification targets using a fully connected layer
Obtain class probabilities with a softmax layer
Step 6: Optimize Model
Compute loss using weighted cross-entropy
Update model using backpropagation and Adam optimizer
Repeat until convergence
Step 7: Generate Output
for each signal S_i **do**
    Assign class with highest probability
end for
return Classification results

In the presented algorithm, FFT preprocessing transforms ECG signals into the frequency domain, enabling dominant frequency identification critical for ECG analysis. The algorithm employs 1D CNNs and residual blocks for feature extraction, adept at capturing temporal patterns and mitigating the vanishing gradient problem, respectively. Vertex infomax pooling constructs a hierarchical graph that preserves topological information, while Graph Convolutional Networks capture dependencies between signals, enhancing feature representation. The READOUT function aggregates these features for classification, with weighted cross-entropy addressing class imbalance during model optimization.

3 Experiments and results analysis

3.1 Datasets Preparation

Our experimental dataset is sourced from the PhysioNet/Computing in Cardiology Challenge 2020 ECG data, referred to as the Challenge Dataset [50]. Given the diverse origins of the ECG recordings in this dataset, it is imperative to standardize the labels across all subjects to ensure consistency in our analysis.

To achieve this, we have implemented three distinct alignment methods, labeled as 14, 24, and 29. Each method represents a unique approach to reconciling variations in labeling conventions and data representation across different sources within the Challenge Dataset. These alignment methods are crucial for harmonizing the data, rendering it suitable for a comprehensive analysis.

Figure 3 depicts a histogram that illustrates the distribution of various ECG categories within the Challenge Dataset after the alignment process. This visualization provides insight into the dataset's diversity and prevalence of different ECG patterns.

3.1.1 CPSC Database and CPSC-Extra Database

The data were collected from China Physiological Signal Challenge(CPSC) including two parts: CPSC Database and CPSC-Extra Database. The CPSC Database contains 6,877 12-lead ECG recordings. The CPSC-Extra Database contains 3,453 12-lead ECG recordings.

All of those recordings were collected from 11 hospitals and sampled at 500 Hz. The duration of the recordings lasts 6 s to 60 s. For the convenience of representation, in subsequent experiments, we will abbreviate these two databases as C database.

3.1.2 PTB Database and PTB-XL Database

The data were collected from Physikalisch Technische Bundesanstalt (PTB), including the PTB and PTB-XL databases. The PTB database contains 549 12-lead ECG recordings, where each recording was sampled at 1000 Hz. The PTB-XL database contains 21,837 ECG recordings of 10-second length with a sampling frequency of 500 Hz. Similarly, we abbreviate these two databases as P databases.

3.1.3 Georgia Database and INCART Database

This section includes two databases: Georgia database and INCART database. The Georgia database was collected from the Southeastern United States. This database

contains 10,344 ECG recordings of 10 s length with a sampling frequency of 500 Hz. The INCART database was collected from the Saint Petersburg Institute of Heart Diseases (INCART). This database contains 32 12-lead ECG recordings, where each recording is 30 min long and contains 12 standard leads and is sampled at 257 Hz. Since the INCART database contains a small number of data samples, we introduce it as a whole with the Georgia database. Similarly, in subsequent experiments, we abbreviate these two databases as G database.

3.2 Experimental Settings

Our proposed model is implemented with Python3.8 language and Pytorch1.9.0 framework. Our experiments are performed on a server with four TITAN RTX graphics cards. The proposed model was trained using Adam optimizer [51]. The initial learning rate is 0.001.

In order to verify the performance of the proposed model, as shown in Table 1 we designed two types of experiments on the Challenge dataset:

3.2.1 Fine-grained classification performance

Many existing methodologies predominantly focus on the coarse-grained classification of arrhythmias, necessitating the identification of 5 or 9 distinct types of arrhythmias. To enhance the alignment with clinical requirements, we have opted for a more granular approach in our study. Specifically, we have categorized all subjects from the Challenge dataset into fine-grained groups, resulting in classifications of 14, 24, and 29 categories, respectively. Subsequently, we evaluated the performance of our proposed model within each of these granular spaces to ensure its robustness and accuracy.

To ensure the reliability of our results, we conducted 20 iterations of fivefold cross-validation, and the results reported are the averages of these iterations.

3.2.2 Cross-center generalization performance

To rigorously evaluate the generalization capabilities of our proposed model, we conducted a series of cross-center validation experiments. In these experiments, the model

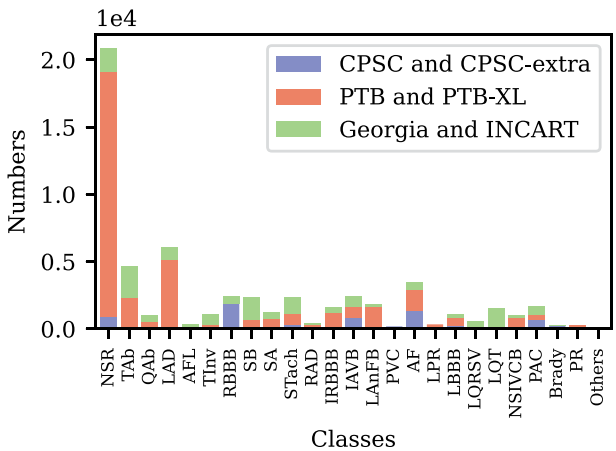


Fig. 3 Histogram of the number distribution of various ECG categories

was trained using data from one center and subsequently tested using data from other centers, ensuring a comprehensive assessment of its performance across diverse data sources. Specifically, the experimental setup is as follows:

1. The model was trained utilizing data from the P and G databases and tested on the C database.
2. The model underwent training on the C database, with subsequent testing on the P and G databases.
3. Training was conducted on the C and G databases, followed by testing on the P database.
4. The model was trained using data from the P database, and tested on the C and G databases.
5. Training was performed on the C and P databases, with testing on the G database.
6. The model was trained on the G database and subjected to testing on the C and P databases.

These meticulously designed experiments aim to provide a robust evaluation of the model's adaptability and effectiveness when applied to data from varying sources, thereby ensuring its reliability and relevance in practical scenarios.

3.3 Evaluation Criterion

In the experiments, we adopt the precision, F1 scores, ROC (Receiver Operating Characteristic curve), AUC (Area Under Curve), and PRC (Precision–Recall curve) [52] to evaluate the performance of different models.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{13}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{14}$$

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

Table 1 Introduction of all experiments for ECG classification

Experiment	Train data	Test data
Challenge-14	Random 80% of all	Random 20% of all
Challenge-24	Random 80% of all	Random 20% of all
Challenge-29	Random 80% of all	Random 20% of all
Cross-C	P + G	C
Cross-PG	C	P + G
Cross-P	C + G	P
Cross-CG	P	C + G
Cross-G	C + P	G
Cross-CP	G	C + P

where TP refers to the number of correctly classified positive samples, FP refers to the number of misclassified positive samples, FN refers to the number of misclassified negative samples, and TN refers to the number of correctly classified negative samples.

3.4 Experiments results

3.4.1 Comparison with Models for ECG Classification

In our comprehensive evaluation, we compared the proposed model's performance with several established models in ECG classification, including Inception1d, LSTM, SE_ECGNet, and XResnet1d. Table 2 compares the performance of our model with other ECG classification models.

Challenge-14: Our model demonstrated robust performance, achieving a precision of 0.820 and an F1 score of 0.740. This is a significant improvement over LSTM (precision: 0.393, F1 score: 0.282), translating to a 108.65% increase in precision and a 162.06% increase in F1 score. Our model slightly surpassed Inception1d (precision: 0.821, F1 score: 0.698), showcasing a 6.00% improvement in F1 score. SE_ECGNet and XResnet1d also showed competitive results but were ultimately outperformed by our model.

Challenge-24: The proposed model achieved the highest F1 score of 0.722, with a precision of 0.753. Inception1d had a decent precision of 0.726 but lagged in F1 score with 0.563, indicating a 28.24% improvement in F1 score by our model. LSTM continued to struggle, with scores significantly lower than the other models, highlighting a 129.82% improvement in F1 score by our model. SE_ECGNet and XResnet1d showed improvements over LSTM but were still outperformed by our model.

Challenge-29: Our model maintained strong performance with a precision of 0.696 and an F1 score of 0.725, outperforming all other models in F1 score. Inception1d had the highest precision at 0.743 but a lower F1 score of 0.542, translating to a 33.76% improvement in F1 score by our model.

Cross-Center Validation: In cross-center validation experiments (Cross-C, Cross-PG, Cross-P, Cross-CG, Cross-G, Cross-CP), our model consistently demonstrated superior F1 scores compared to the other models, showcasing its robustness and generalization capabilities across different data centers.

On average, our proposed model achieved a precision of 0.586 and an F1 score of 0.531, outperforming all other models in terms of F1 score. Inception1d showed a similar average precision but a lower F1 score, indicating a potential issue in balancing precision and recall, and a 14.97% improvement in F1 score by our model. LSTM had the lowest average scores, highlighting challenges with long ECG recordings and a 208.72% improvement in F1 score by our model. SE_ECGNet and XResnet1d showed improvements over LSTM but were still outperformed by our model.

Figures 4, 5 and Table 3a further reveal the high performance of our model in the challenge-14 experiment, in the challenge-24 experiment and the challenge-29 experiment. Figure 5 shows the confusion matrix of eight representative types of ECG abnormalities. For example, it is important to detect atrial fibrillation (AF) for planning the appropriate treatment; in fact, the confusion matrix of atrial fibrillation (AF) in three experiments shows that our method can accurate detect the AF.

Table 2 Comparison with models commonly used for ECG classification

Experiment	Inception1d [26]		LSTM [20]		SE_ECGNet [53]		XResnet1d [54]		Proposed	
	Precision	F1 scores	Precision	F1 scores	Precision	F1 scores	Precision	F1 scores	Precision	F1 scores
Challenge-14	**0.821**	0.698	0.393	0.282	0.715	0.645	0.798	0.712	0.820	**0.740**
Challenge-24	0.726	0.563	0.404	0.312	0.655	0.461	0.724	0.614	**0.753**	**0.722**
Challenge-29	**0.743**	0.542	0.347	0.200	0.645	0.450	0.742	0.566	0.696	**0.725**
Cross-C	0.516	0.447	0.200	0.111	0.480	0.396	0.553	0.505	**0.628**	**0.531**
Cross-PG	**0.636**	0.514	0.238	0.153	0.617	0.460	0.423	0.321	0.564	**0.534**
Cross-P	**0.532**	0.410	0.216	0.145	0.474	0.401	0.495	0.414	0.493	**0.449**
Cross-CG	**0.499**	0.363	0.172	0.129	0.484	0.360	0.480	0.359	0.461	**0.414**
Cross-G	0.465	0.304	0.203	0.113	0.459	0.311	0.444	0.310	**0.488**	**0.330**
Cross-CP	**0.387**	0.315	0.137	0.103	0.372	0.287	0.389	0.307	0.367	**0.334**
Average	**0.592**	0.462	0.257	0.172	0.545	0.419	0.561	0.456	0.586	**0.531**

Bold indicates the best-performing indicator across all tested parameters

These results collectively underscore the effectiveness of our proposed model in ECG classification, demonstrating its ability to achieve a balanced trade-off between precision and recall, and its robustness across various experimental settings.

3.4.2 Classification Performance of Proposed Model and Ablation Models

To assess the effectiveness of the proposed model, we conducted a comprehensive evaluation, comparing its performance with three ablation models: 1D-ResNet34, 2-ResGCNet (incorporating a bidirectional mechanism and a 2-layer GCN), and 6-ResGCNet (integrating a bidirectional mechanism and a 6-layer GCN). The 1D-ResNet34 model, consisting solely of the bidirectional feature extraction mechanism for intra-class dependency learning, validates the enhanced discriminative capability imparted by our model's bidirectional mechanism. On the other hand, the 2-ResGCNet and 6-ResGCNet models were introduced to demonstrate that the strength of our model is attributed to the VIP pooling strategy rather than the number of GCN layers employed. The evaluation is systematically categorized into graph learning performance and generalization capabilities of the models.

Graph Learning Performance: In the evaluation of graph learning performance across various datasets, the proposed model consistently outshone the ablation models, including 1D-ResNet34, 2-ResGCNet, and 6-ResGCNet. Figure 6 shows the ROC and AUC value of our model for each type of ECG anomaly in the challenge-24 experiment. Our model achieved the AUC values ≥ 0.85 for all types of ECG anomaly. This shows that our method can distinguish these main categories of ECG anomaly very well. For some types of ECG anomaly that are harmful, such as left bundle branch block(LBBB), our model achieved AUC values of 0.98. These results show the effectiveness of our method for identify all categories of ECG anomaly. Table 3a shows the performance of proposed model and ablation models. Specifically, in the Challenge-14 dataset, the proposed model exhibited a 5.81% increase in precision and a 5.56% increase in F1 score compared to 1D-ResNet34. Moving on to the Challenge-24 dataset, the model continued

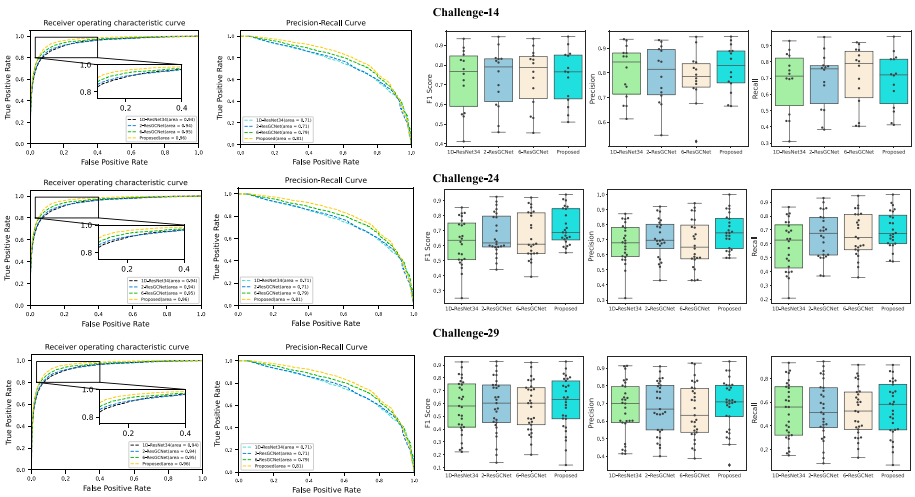


Fig. 4 Performance of proposed model and ablation models on challenge-14, challenge-24, and challenge-29 experiments

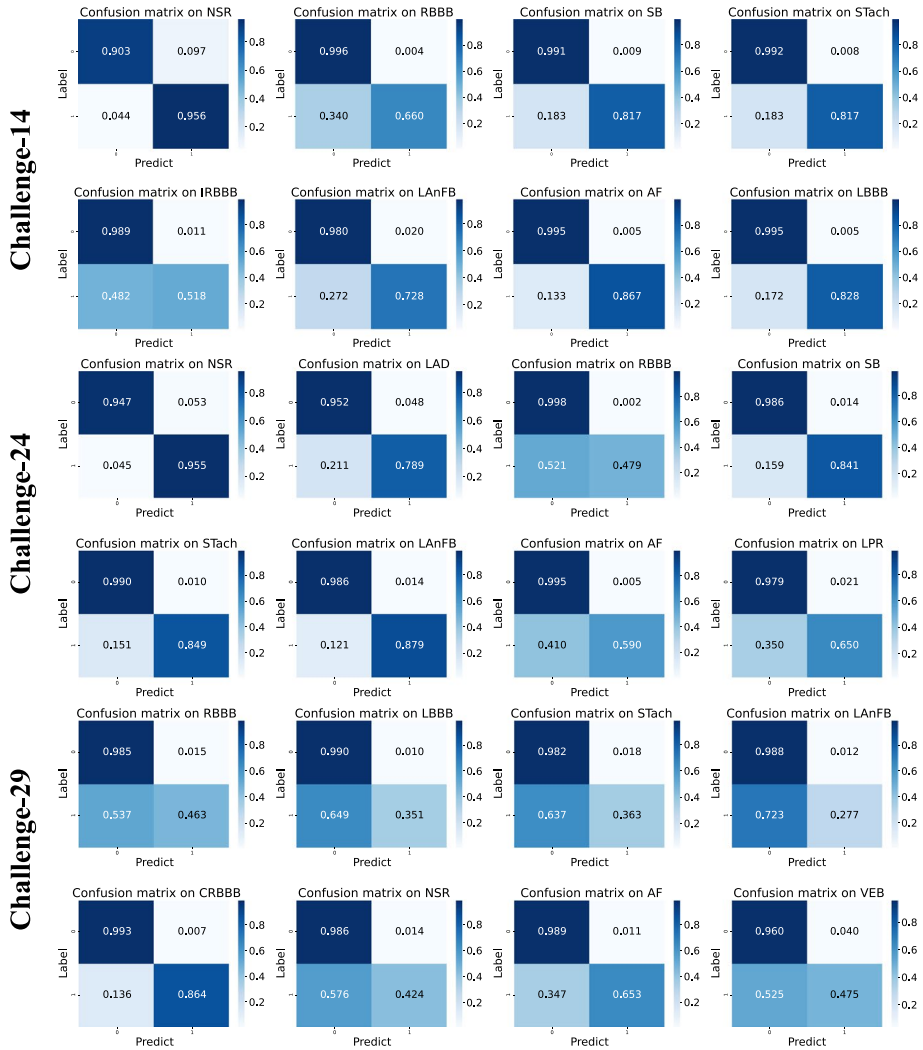


Fig. 5 Confusion matrix of proposed model on challenge-14, challenge-24, and challenge-29 experiments. This shows the confusion matrix of the proposed model in some ECG labels

to demonstrate its efficacy, outperforming 1D-ResNet34 by 4.29% in precision and a remarkable 15.89% in F1 score. In the Challenge-29 dataset, the proposed model maintained its superior performance, showing a 4.04% improvement in precision and a 3.87% improvement in F1 score over 1D-ResNet34. On average, the proposed model achieved a 1.21% increase in precision and an 8.18% increase in F1 score across all datasets when compared to 1D-ResNet34, highlighting its robustness and effectiveness in ECG abnormality classification.

Generalization Performance: In evaluating generalization performance across various experimental conditions, including Cross-C, Cross-PG, Cross-P, Cross-CG, Cross-G, and Cross-CP, the proposed model demonstrated remarkable robustness and consistency. As detailed in Table 3b, the results reveal that the proposed model achieved an average precision of 0.500 and an F1 score of 0.432, outperforming the ablation models in most scenarios.

Table 3 Classification performance of proposed model and ablation models

Experiment	1D-ResNet34		2-ResGCNet		6-ResGCNet		Proposed	
	Precision	F1 scores	Precision	F1 scores	Precision	F1 scores	Precision	F1 scores
(a) Experiments on fine-grained classification performance								
Challenge-14	0.775	0.701	0.790	0.710	0.796	0.730	**0.820**	**0.740**
Challenge-24	0.796	0.623	0.707	0.675	0.673	0.665	**0.753**	**0.722**
Challenge-29	0.669	0.698	0.677	0.706	0.657	0.697	**0.696**	**0.725**
Average	0.747	0.674	0.725	0.697	0.709	0.697	**0.756**	**0.729**
(b) Experiments on generalization performance								
Cross-C	0.521	0.285	0.528	0.303	0.524	0.280	**0.628**	**0.531**
Cross-PG	0.601	0.469	0.603	0.284	**0.625**	0.294	0.564	**0.534**
Cross-P	0.476	0.428	0.485	0.411	0.476	0.440	**0.493**	**0.449**
Cross-CG	**0.464**	0.363	0.427	0.388	0.443	0.384	0.461	**0.414**
Cross-G	0.449	0.309	0.478	0.310	0.419	0.293	**0.488**	**0.330**
Cross-CP	0.353	0.332	0.358	0.331	0.361	0.319	**0.367**	**0.334**
Average	0.477	0.364	0.480	0.338	0.475	0.335	**0.500**	**0.432**

Bold indicates the best-performing indicator across all tested parameters

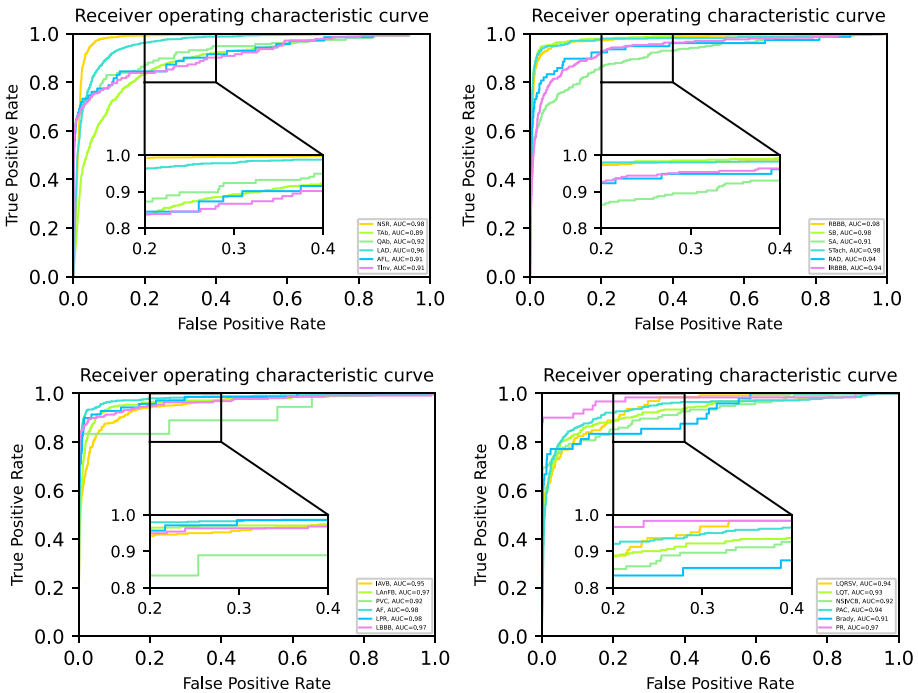


Fig. 6 Receiver operating characteristic curves (ROC) of the proposed model on Challenge-24 experiment

Particularly noteworthy is the model's performance in the Cross-C experiment, where it achieved a precision of 0.628 and an F1 score of 0.531, showcasing a significant improvement in its ability to generalize across different data centers. In the Cross-PG and Cross-P experiments, the model maintained high performance with precision scores of 0.564 and 0.493 and F1 scores of 0.534 and 0.449, respectively. These results underscore the model's commendable generalization capabilities, further solidifying its

effectiveness across diverse and challenging experimental conditions in ECG abnormality classification.

In addition to the overall performance of the models, it is critical to achieve a good performance for various anomalities. Figure 7 shows the classification performance of the model for some labels in Challenge-24 experiment. Our proposed model almost shows best performance in F1. Compared with other models, this shows that proposed model is robust to predict various ECG labels. However, the precision and recall of proposed model prediction cannot all surpass other models. For example, when predicting RBBB, SE_ECGNet achieves the best performance in precision. When predicting SB, 6-ResGCNet achieves the best performance in recall. The proposed model achieves robust performance in the balance between precision and recall.

3.4.3 Comparative Analysis of Graph Convolution Techniques in ECG Classification

In the section, we present a detailed comparison of our algorithm's performance against existing graph convolution methodologies applied to ECG signal classification. This comparison, as is shown in Table 4, showcases our approach's superiority, notably achieving an F1 score of 0.729. We attribute this success to our novel utilization of node relationships and graph pooling, areas previously underexplored in related research. Our findings suggest significant potential for enhancing ECG classification accuracy by integrating these advanced graph-based strategies, marking a notable advancement in the domain.

4 Discussion

In our comprehensive evaluation, the proposed model shows exhibited superior performance in ECG classification, as evidenced by the quantitative results and the visual insights provided by t-SNE, a popular nonlinear dimensionality reduction technique. t-SNE has played a pivotal role in our analysis, ensuring that similar data points in the high-dimensional space are brought closer in the low-dimensional space, facilitating a clear visual interpretation of the model's feature extraction capabilities.

Figure 8 in our results showcases the t-SNE visualizations for various ECG categories, providing a visual testament to the effectiveness of our model. Compared to other models, the features extracted by our proposed model are distinctly separable, highlighting its robustness in distinguishing between different ECG abnormalities. This visual confirmation aligns with our quantitative findings, where the model has shown improvements in precision and F1 scores, surpassing other models and ablation studies by up to 5.8% and 5.5%, respectively, in certain experiments.

However, it is crucial to acknowledge the limitations of our current approach. The model does not account for the spatial relationships between ECG leads or the clinical correlations between different ECG abnormalities, vital for a holistic understanding of cardiac conditions. Addressing data imbalance, especially for rare ECG abnormalities, remains a challenge, as the model's performance is contingent on the availability of substantial data for feature extraction.

In our future work, we aim to address these limitations by integrating unsupervised learning into ECG multi-label classification and incorporating spatial relationships

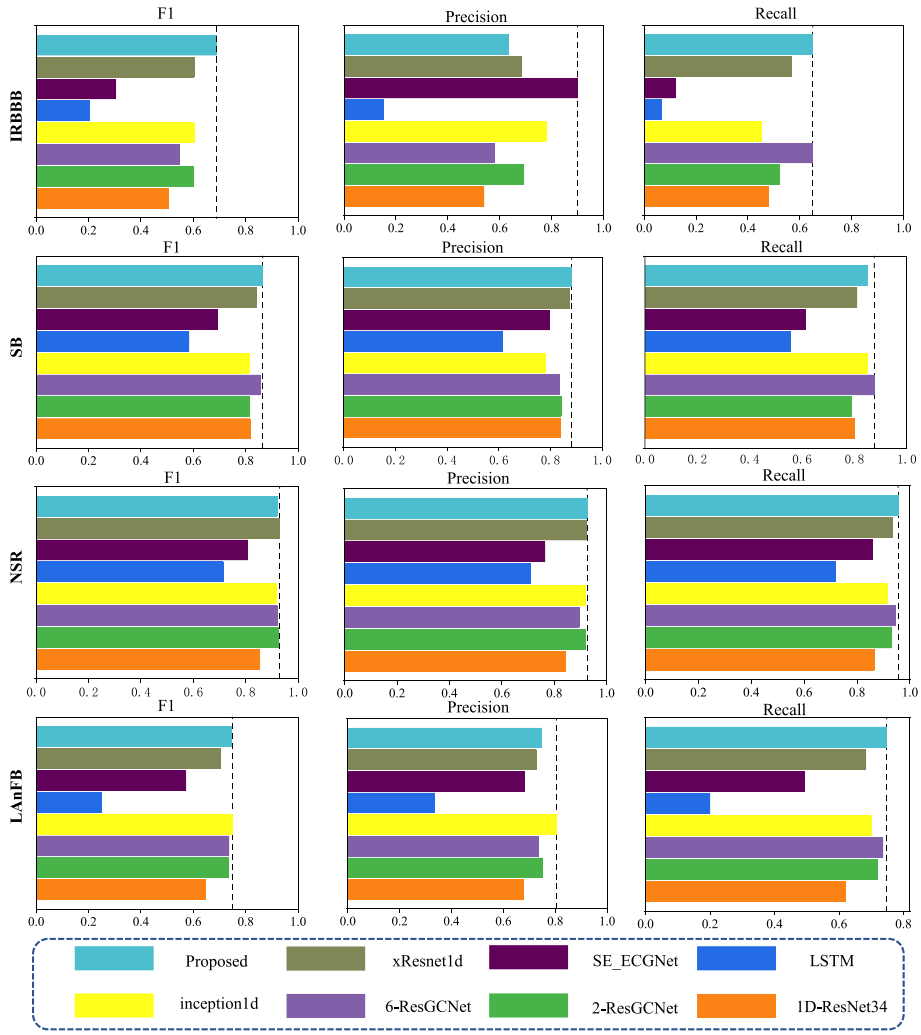


Fig. 7 Comparison of the prediction performance of models on the Challenge-24 experiment. This is a brief comparison of some ECG labels, such as IRBBB, SB, NSR, and LAnFB

Table 4 Comparative performance of graph convolution methods for ECG classification

	Res-Blocks+Bi-GRU+GCN [55]	GAT-based method [56]	MLWGAT [56]	Proposed
F1	0.603	0.448	0.440	**0.729**

Bold indicates the best-performing indicator across all tested parameters

and clinical correlations into our model. This holistic approach is expected to enhance the classification performance, especially for ECG abnormalities with limited data, and contribute to advancing precision medicine in cardiac care.

In summary, while our proposed model has demonstrated promising results in ECG classification, bolstered by visual validations through t-SNE, there is a clear path for future improvements. Addressing the current limitations and integrating advanced learning techniques will be crucial in our ongoing efforts to enhance the model's performance and contribute to the field of cardiac anomaly detection.

5 Conclusion

In this study, we have introduced ECGNet, an end-to-end model designed to classify ECG abnormalities in 12-lead ECG recordings, setting a new benchmark in automated arrhythmia diagnosis. Distinct from previous research, our approach eliminates the need for intricate ECG preprocessing, such as converting raw ECG data into individual heartbeat information, preserving the integrity of the original data. This methodology enables the extraction of a more comprehensive set of features directly from the ECG; utilizing information from all leads to enhance the classification performance.

The architecture of ECGNet is further augmented by the integration of multi-level graph learning, a unique feature not commonly found in existing deep learning-based ECG research. This addition refines the features associated with ECG abnormalities, significantly enhancing classification performance, as evidenced by our extensive experiments on the Challenge Dataset. The proposed model has demonstrated superior performance compared to commonly used ECG classification models and robust generalization capabilities in cross-database experiments, showcasing its reliability and effectiveness in various scenarios.

However, we acknowledge the limitations of our work, particularly in terms of addressing data imbalance and incorporating spatial relationships between ECG leads and correlations between different ECG abnormalities. Future work will focus on these areas, with plans to integrate unsupervised learning and explore novel methods to further enhance the model's performance, especially for ECG abnormalities with limited available data.

In conclusion, ECGNet represents a significant advancement in the field of ECG classification, delivering robust performance in the multi-label classification of ECG abnormalities. The integration of multi-level graph learning and the ability to directly process raw ECG data have positioned ECGNet as a powerful tool for aiding clinicians in accurately and timely diagnosing cardiac abnormalities. The promising results and the clear

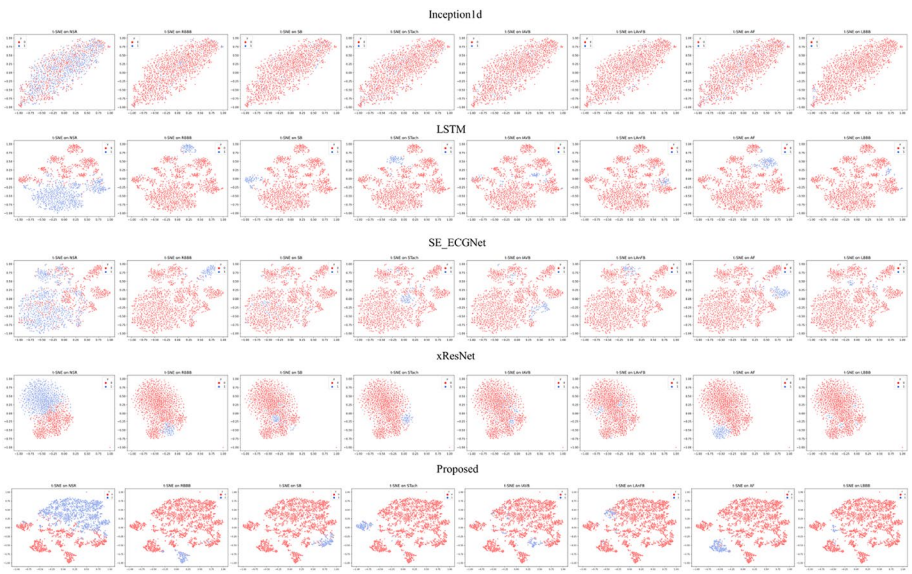


Fig. 8 T-SNE visualization of the learned representation of models on Challenge-24 experiment

visual validation of the model's effectiveness pave the way for future advancements and applications in ECG analysis.

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